

MSTAR PRACTICE WORKSHEET**Lesson 7-9: Reflect Points Across Axes**

Grade 6 Mathematics · Standard 6.NOS.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: D**What is the reflection of $(4, -3)$ over the x-axis?

- ☐ (A) $(-4, -3)$
- ☐ (B) $(-4, 3)$
- ☐ (C) $(3, -4)$
- ☒ (D) $(4, 3)$

Reflecting over the x-axis changes the sign of the y-coordinate. The x stays the same (4), and the y changes from -3 to 3 . The answer is $(4, 3)$.

Question 2 **SELECTED RESPONSE — CORRECT: B**What is the reflection of $(-2, 6)$ over the y-axis?

- ☐ (A) $(-2, -6)$
- ☒ (B) $(2, 6)$
- ☐ (C) $(2, -6)$
- ☐ (D) $(6, -2)$

Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -2 to 2 , and y stays 6 . The answer is $(2, 6)$.

Question 3 SELECTED RESPONSE — CORRECT: B

Which point is on the y-axis?

- ☐ A (-2, 4)
- ☒ B (0, 5)
- ☐ C (3, 3)
- ☐ D (5, 0)

Points on the y-axis have x-coordinate = 0. (0, 5) is on the y-axis.

Question 4 SELECTED RESPONSE — CORRECT: C

Point A is at (-3, 5). It is reflected over the x-axis to A', then A' is reflected over the y-axis to A''. What are the coordinates of A''?

- ☐ A (-3, -5)
- ☐ B (-3, 5)
- ☒ C (3, -5)
- ☐ D (3, 5)

Over x-axis: (-3, 5) → (-3, -5). Then over y-axis: (-3, -5) → (3, -5). Both coordinates changed sign.

Question 5 SELECTED RESPONSE — CORRECT: D

Captain Vega folds the map along the x-axis. A marker at (7, 2) lands on its mirror image. What are the new coordinates?

- ☐ A (-7, 2)
- ☐ B (-7, -2)
- ☐ C (2, 7)
- ☒ D (7, -2)

Folding along (reflecting across) the x-axis flips the sign of the y-coordinate only. (7, 2) becomes (7, -2).

Question 6 **SELECTED RESPONSE — CORRECT: C**

A buried chest is marked at $(-5, 3)$. Vega folds the map along the y -axis to find its mirror twin. Where is the twin?

- ☐ (A) $(-5, -3)$
- ☐ (B) $(3, -5)$
- ☒ (C) $(5, 3)$
- ☐ (D) $(5, -3)$

Reflecting across the y -axis flips the sign of the x -coordinate only. $(-5, 3)$ becomes $(5, 3)$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 **TWO-PART QUESTION****Part A — correct answer: B**

Reflect the point $(5, -2)$ across the x -axis. What are the coordinates of the image?

- ☐ (A) $(-2, 5)$
- ☒ (B) $(5, 2)$
- ☐ (C) $(-5, -2)$
- ☐ (D) $(-5, 2)$

Reflecting across the x -axis changes the sign of the y -coordinate only: $(x, y) \rightarrow (x, -y)$, so $(5, -2) \rightarrow (5, 2)$.

- **A:** The coordinates were swapped. Keep x at 5 and change the sign of y .
- **C:** That reflects across the y -axis, which changes x . Reflecting across the x -axis changes y .
- **D:** Both signs were changed. Reflecting across the x -axis changes only the y -coordinate.

Part B — correct answer: B

Why does reflecting across the x -axis change only the y -coordinate?

- ☐ (A) The x -coordinate is always positive after a reflection.
- ☒ (B) The x -axis is horizontal, so the point flips vertically while its horizontal position stays fixed.
- ☐ (C) Only the second coordinate can be negative.
- ☐ (D) Reflections always change exactly one coordinate to zero.

A reflection across a line flips the point to the other side of that line at the same distance. Across the horizontal x -axis, only the vertical position changes.

Question 8

WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Reflecting Across the Wrong Axis

Dante reflected $(3, 7)$ across the y -axis. He wrote: 'Change the sign of y , so the image is $(3, -7)$.'

Explain Dante's misconception and give the correct image point.

Model answer: $(-3, 7)$. Reflecting across the y -axis (a vertical line) flips the point horizontally: $(x, y) \rightarrow (-x, y)$. Dante used the x -axis rule, $(x, y) \rightarrow (x, -y)$, which reflects across the horizontal axis instead.

2 points	Student identifies that a reflection across the y -axis changes the sign of the x -coordinate, gives $(-3, 7)$, and explains that Dante applied the x -axis rule instead.
1 point	Student gives $(-3, 7)$ without explaining which rule Dante used.
0 points	Student agrees with Dante or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A triangle has vertices at $A(1, 2)$, $B(4, 2)$, and $C(4, 6)$. If you reflect the entire triangle over the y -axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?

Model answer: Reflect each vertex over the y -axis by flipping only the x -coordinate: $A(1, 2) \rightarrow A'(-1, 2)$, $B(4, 2) \rightarrow B'(-4, 2)$, and $C(4, 6) \rightarrow C'(-4, 6)$. The reflected triangle is congruent to the original — same side lengths and same angles — because a reflection is a mirror flip that turns the figure around without stretching or shrinking it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.